Result

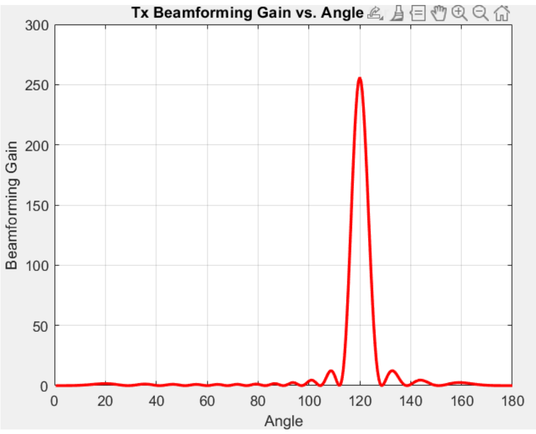
Task1:

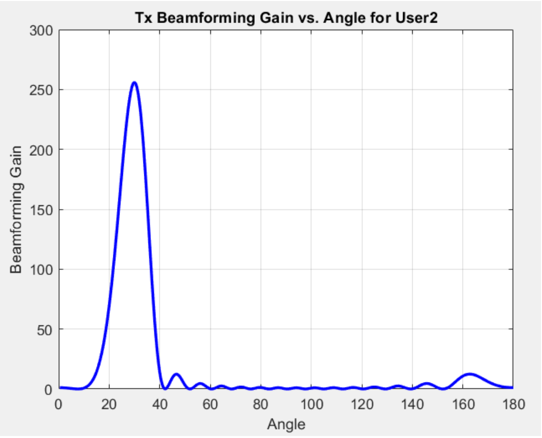
Q1:

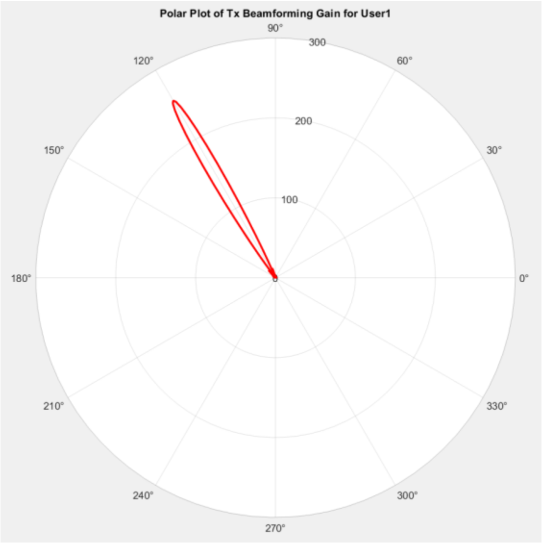
```
Task 1: Calculate AoD
Rx1 Actual Angle: 115.595208 degrees
Rx2 Actual Angle: 25.076622 degrees
Rx1 Optimal beam: 120.000000 degrees
Rx2 Optimal beam: 30.000000 degrees
```

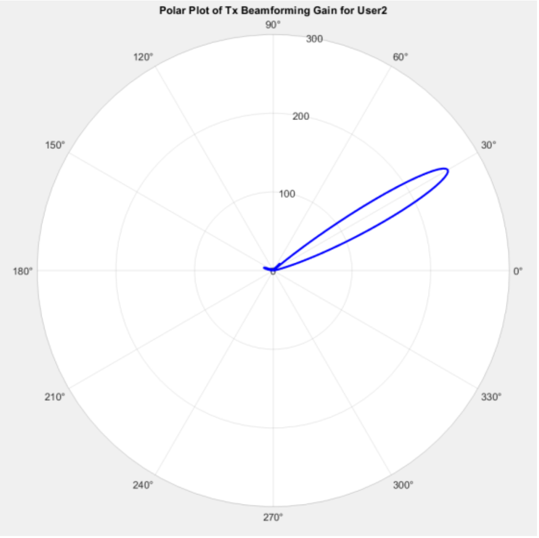
會發現最好的角度會是最靠近實際角度的十的倍數

Q2:









會發現 tx_gain 最高的地方是在 beam 打出去的角度上，之後會往兩側高速下降

Task2:

Q1:

```
Distance  = 23.382181 m
Tx_gain   = 81.693222
Rx1 power = -48.307848 dBm
Rx1 SNR   = 39.692152 dB
```

Task3:

Q1:

```
User1 SINR
Signal Power (User1) = -48.307848 dBm
Rx1 interference power = -67.134186 dBm
Rx1 SINR = 18.826338 dB
```

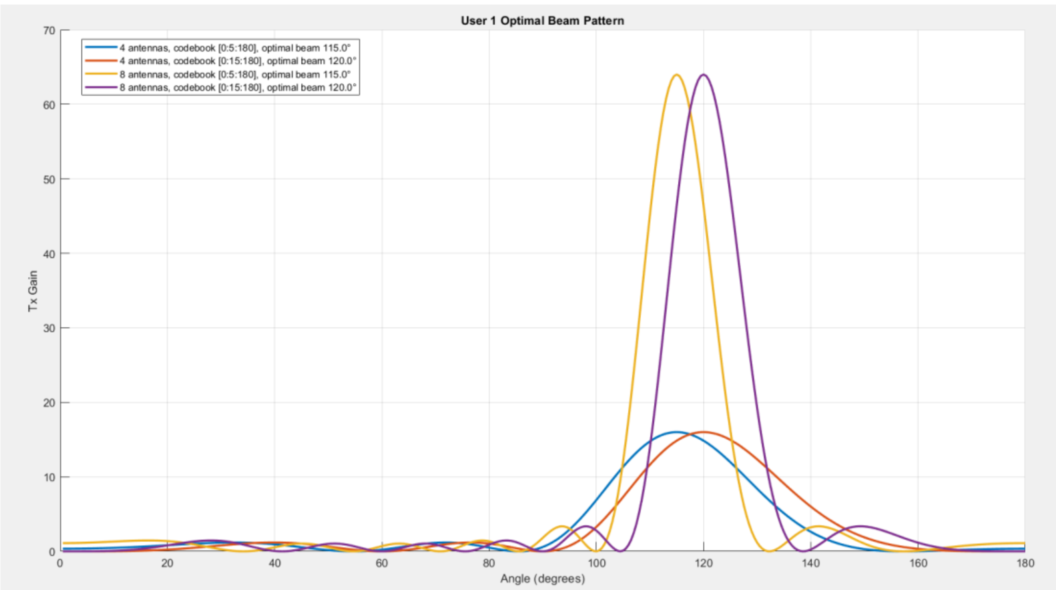
可以發現 SNR 跟 SINR 只有差距有 20db，代表 user2 的 side lobe 的影響對 user1 非常大。

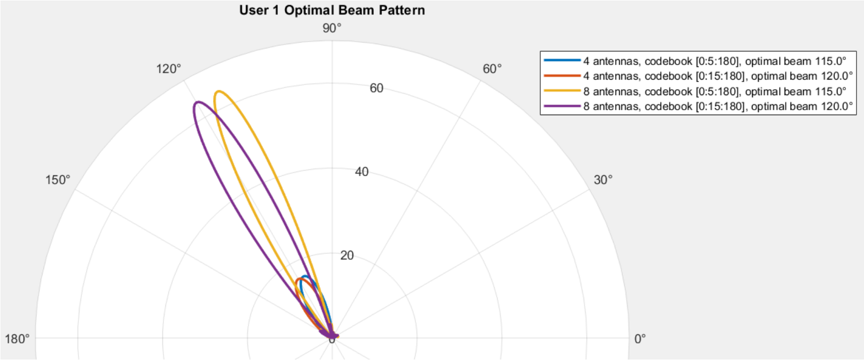
Comparison:

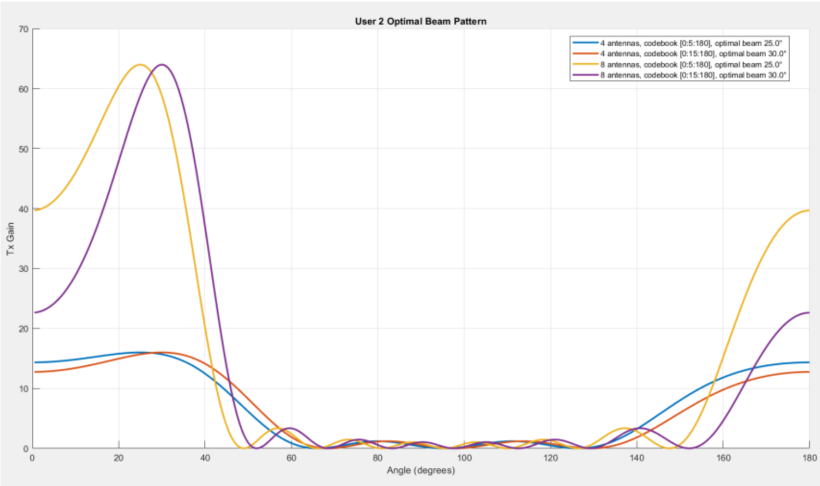
Plots of the optimal beam θ* for user 1 and 2 under different antenna number (4 or 8) and codebook size ([0:5:180] or [0:15:180]):

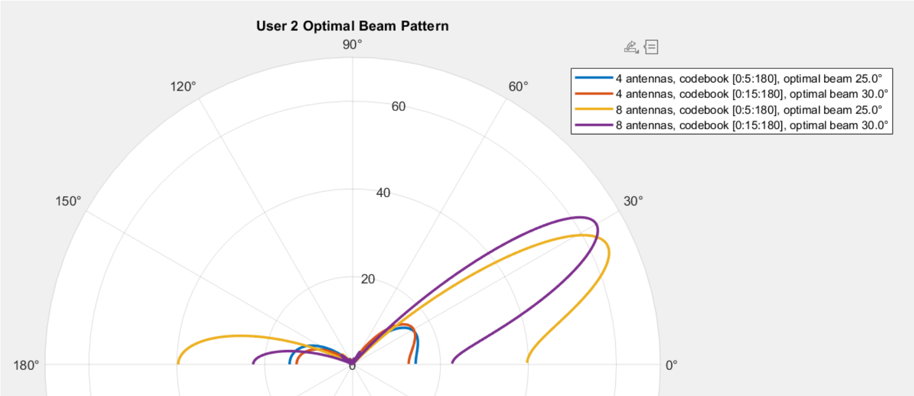
我先放 user1(前兩張圖)再放 user2

可以發現依據 code size 的不同 tx_gain 最大值的角度也會不同









Answers from 20 random runs for tasks 1 to 3:

```
Averaged Results over 20 runs:
Average Rx1 power              = -39.546448 dBm
Average Rx1 SNR                = 48.453552 dB
Average Rx1 interference pwr = -65.966759 dBm
Average Rx1 SINR               = 26.420311 dB
```

與原本的 SNR 差了約 10db，隨機 20 次下會有比較多與 beam 對齊比較好的情形，使平均升高。